



The Impact of Working Memory Training on Cognitive Reappraisal Ability Among Older Women

Qing Chai¹ · Yue Gao¹ · Yajie Gong¹ · Xin Zhao¹ 

Received: 16 July 2025 / Accepted: 2 March 2026
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2026

Abstract

While working memory training (WMT) has demonstrated promise in mitigating cognitive decline among older adults, its efficacy in addressing age-related challenges in emotion regulation—specifically through cognitive reappraisal (CR)—remains underexplored, particularly in older women. In this research, we conducted two studies to investigate (a) the age-moderated relationship between working memory (WM) updating and CR, and (b) whether WMT could enhance CR in this demographic. In Study I, we assessed WM through a numerical updating task and a 1-back task, as well as CR through an emotion regulation task, involving 101 young women (aged 18–30) and 114 older women (aged 65–85). Study II included 69 older women (aged 65–85), who were randomly assigned to a Training group ($n=22$), an Active Control group ($n=22$), or a No-contact control group ($n=25$), with the Training group completing 20 sessions of adaptive training. Results from Study I indicated that young women outperformed older women on both WM and CR tasks, and age significantly moderated their relationship; a positive WM-CR correlation was identified exclusively in young women. In Study II, only the training group demonstrated significant post-intervention improvements in both WM updating and CR performance compared to the control groups. We conclude that WMT can effectively enhance WM and, crucially, improve CR abilities in older women, thereby positioning it as a viable intervention for age-related emotional dysregulation.

Keywords Working memory training · Cognitive reappraisal · Older women

Introduction

The ability to engage in adaptive emotion regulation represents a cornerstone of mental health and psychological resilience across the lifespan. Within the repertoire of emotion regulation strategies, cognitive reappraisal (CR)—defined as the cognitive reinterpretation of the meaning of an emotionally salient stimulus to alter its affective impact—is consistently identified as a highly effective and cognitively sophisticated strategy (Gross, 1998; Urry & Gross, 2010). However, its efficacy is contingent upon the availability of adequate cognitive resources. Convergent empirical and neurobiological evidence places working

memory (WM), particularly its updating function, at the core of this cognitive resource pool. WM updating facilitates the active maintenance, manipulation, and revision of goal-relevant information (Miyake et al., 2000), processes that are functionally analogous to the generation and maintenance of alternative appraisals central to cognitive reappraisal (CR). Neuroimaging studies corroborate this mechanistic overlap by revealing co-activation within the dorsolateral prefrontal cortex (DLPFC), a region integral to cognitive control, during both WM updating and CR tasks (Ochsner et al., 2012). Consequently, individual differences in WM are consistently linked to the effectiveness of reappraisal efforts in experimental paradigms (Schmeichel & Tang, 2015; Sperduti et al., 2017). Emotion regulation is fundamentally a cognitive control process (Diestel & Schmidt, 2011). In everyday life, individuals often need to reduce negative emotions (e.g., failure in exams); cognitive reappraisal refreshes the meaning of negative situations, enabling individuals to reinterpret them in a more constructive way (Ochsner et al., 2012). Specifically, refreshing refers to transforming negative cognitions into more

Qing Chai and Yue Gao contributed equally to this work.

✉ Xin Zhao
psyzhaoxin@nwnu.edu.cn

¹ School of Psychology, Northwest Normal University, 967 Anning East Road, Lanzhou City, Gansu Province 730070, China

neutral or positive content (Ochsner et al., 2012). In this process, individuals must activate and utilize positive or neutral representations in working memory, interrupt the connections between negative representations, and replace them with new content to alleviate negative emotions; all these processes rely on the ability to refresh (Levens & Gotlib, 2010).

A critical, yet unresolved, question concerns the stability of the WM-CR relationship in the context of normative aging. This period is characterized by a well-documented trajectory of decline in the efficiency of fluid cognitive abilities, including WM, primarily attributed to alterations in prefrontal circuitry (Vanderhasselt et al., 2021). Paradoxically, this neurocognitive decline does not uniformly extend to the emotional domain; emotional well-being often remains stable, and the ability to implement certain emotion regulation strategies, such as CR, may be preserved or even enhanced in later life (Charles & Carstensen, 2008; Urry & Gross, 2010). This dissociation suggests potential adaptive shifts, wherein older adults may increasingly leverage crystallized knowledge, life experience, and motivational changes to support emotion regulation, potentially compensating for declines in fluid cognitive resources. This compensatory perspective, aligned with models such as Selective Optimization with Compensation (SOC) applied to emotion regulation (Urry & Gross, 2010), introduces a theoretical quandary: if CR can be maintained despite WM decline, does the foundational reliance of CR on WM updating persist into late adulthood, or is it superseded by alternative, experience-dependent mechanisms?

To reconcile this paradox, we advance an efficiency bottleneck hypothesis (Growney & English, 2023). This framework posits that WM updating remains a necessary cognitive substrate for the effortful, online operations of CR. However, age-related reductions in WM processing efficiency may lower its functional capacity below a critical threshold required for measurable support in standardized, laboratory-based CR tasks (Martins et al. 2018a; 2018b). When operating below this threshold, the statistical association between WM and CR performance may be attenuated or absent, creating a performance bottleneck that masks the underlying resource dependency. The apparent preservation of CR in ecological contexts may thus reflect a strategic reliance on more automatic, experience-based regulatory approaches that do not fully engage the overtaxed updating system. A key, testable implication of this hypothesis is that remediating the bottleneck—by selectively enhancing WM updating efficiency—should restore its measurable contribution and lead to concomitant gains in CR ability.

Adaptive WMT provides an ideal paradigm for empirically evaluating this proposition. Substantial evidence confirms the plasticity of WM networks, demonstrating that WMT, which dynamically adjusts task difficulty to maintain an optimal level of challenge, can produce sustained improvements in WM performance and transfer effects to related cognitive domains (McNab et al., 2009; Szymkowicz et al., 2022). Pertinently, preliminary studies indicate that WMT can augment CR proficiency in younger adult and clinical samples (Long et al., 2024; Peng et al., 2019). Extending these findings to healthy older adults is crucial, yet existing evidence remains inconsistent, with some interventions failing to demonstrate transfer to emotion regulation (Vanderhasselt et al., 2021). We postulate that such null findings may arise from training protocols of insufficient intensity or adaptability to elevate WM efficiency beyond the putative critical threshold for this population, or from insufficient targeting of demographic cohorts for whom the WM-CR link is most salient.

This investigation, therefore, focuses specifically on older women. This focus is motivated by a confluence of factors: women exhibit a higher lifetime prevalence of disorders characterized by emotion dysregulation (Holden, 2005), display sexually dimorphic neural activation patterns during emotion regulation tasks (McRae et al., 2008), and emerging data suggest that the association between WM updating and CR may be particularly robust in females (Zhao et al., 2023). Consequently, older women represent a strategic population in which to probe the plasticity of the WM-CR link, as they constitute a group for whom the targeted cognitive mechanism is both susceptible to age-related decline and theoretically central to emotional health.

To address these interconnected questions, we conducted a two-study investigation. Study I employed a cross-sectional design to examine whether age moderates the relationship between WM updating ability and CR performance. We hypothesized that the positive correlation observed in young women would be absent or significantly attenuated in older women, consistent with the presence of an efficiency bottleneck. Building on these findings, Study II adopted a randomized, controlled experimental design to assess the causal efficacy of WMT in enhancing CR in older women. We predicted that participants receiving WMT would exhibit significant improvements in both WM updating and CR compared to those in fixed-difficulty active control or no-contact control conditions. Furthermore, we anticipated that post-intervention improvements in WM would be strongly correlated with enhancements in CR, providing direct evidence for the re-emergence of the latent cognitive-emotional link once the processing bottleneck is alleviated.

Study I

In Study I, we employed a cross-sectional design to examine the association between WM updating and CR ability and to determine whether this relationship was moderated by age. The sample comprised 101 young adult women (age range=18–30 years) and 114 older adult women (age range=65–85 years). All participants completed a battery of computerized tasks assessing WM updating—specifically, a numerical WM task and a 1-back task—followed by a well-validated emotion regulation task designed to assess CR efficacy. Based on the well-documented age-related declines in both executive function and the efficiency of effortful emotion regulation strategies, we hypothesized that older women would exhibit significantly poorer performance on measures of WM updating and CR compared to their younger counterparts. Furthermore, we predicted that age would function as a significant moderator of the WM–CR link, with a strong positive correlation anticipated in the young adult group and an attenuated or non-significant association expected in the older adult group, potentially reflecting an age-related efficiency bottleneck in resource-dependent emotional regulation.

Method

Participants

Participants were recruited via online advertisements and community posters, comprising 101 young women ($M=21.36$, $SD=2.80$; age range=18–32) and 114 older women ($M=70.56$, $SD=5.27$; age range=65–85). All participants reported no history of mental illness and scored within the normal range for anxiety and depression. Older women also scored ≥ 25 on the Mini-Mental State Examination (MMSE; Folstein et al., 1975), indicating intact cognitive function. All participants were right-handed and had normal or corrected-to-normal vision. Independent t -tests revealed significantly lower anxiety ($t(213)=5.08$, $p<.001$) and depression ($t(213)=8.42$, $p<.001$) scores in the older subgroup (65–85 years) compared to young women. Consequently, years of education, anxiety, and depression scores were included as covariates in subsequent analyses. A priori power analysis using G*Power 3.1 (Faul et al., 2007) for linear multiple regression (fixed model, R^2 increase) with an effect size of $f^2 = 0.15$, $\alpha=0.05$, $1-\beta=0.80$, indicated a minimum sample size of 55 (1 tested predictor, 6 total predictors). The final sample of 215 participants exceeded this threshold. All participants provided written informed consent prior to participation and were compensated for their

involvement after the study. The study was approved by the Ethics Committee of Northwest Normal University and was performed in accordance with the approved guidelines.

Materials

The WM tasks included a numerical WM task (Zhao et al., 2019) and a 1-back task (Yang et al., 2011). CR was assessed using an emotion-regulation task (McRae et al., 2012; Zhao et al., 2023).

Numerical WM Task This task assessed WM-updating ability under two levels of difficulty (Broadway & Engle, 2010). Digits from 0 to 9 were presented sequentially at either 1750 ms (easy) or 750 ms (difficult) intervals. Each trial began with a randomly selected digit and included sequences of 5, 7, 9, or 11 digits. Participants were instructed to recall the last three digits in sequence after the final stimulus. Practice trials were completed prior to the main task, with a $\geq 50\%$ accuracy requirement to proceed. Task accuracy was recorded as the dependent variable.

1-Back Task In this task, participants compared each digit to the one immediately preceding it, pressing “F” for matches and “J” for non-matches. A 20-trial practice phase required $\geq 80\%$ accuracy. The formal task consisted of three blocks of 84 trials each. Each trial began with a 1500 ms fixation cross, followed by a 500 ms digit and a 2500 ms response window. Breaks were provided between blocks. Performance was indexed by accuracy, with higher ratios indicating poorer updating ability (Zhao et al., 2018). The total task duration was approximately 8 min.

Emotion Regulation Task CR was assessed using an emotion-regulation task, adapted from McRae et al. (2012) and Zhao et al. (2023). Each trial presented an instruction (“watch” or “reappraise”) for 3000 ms, followed by a fixation point (300 ms), and then a neutral or negative image (2000 ms) from the International or Chinese Affective Picture Systems (Lang et al., 1999; Lu et al., 2005). Participants rated the emotional intensity of the image (1=not at all; 9=very negative) within a 4000 ms window, followed by a 2000 ms blank screen. The task comprised three blocks (25 trials each): (1) watch-neutral, (2) watch-negative, and (3) reappraise-negative. Instructions encouraged participants to either observe the image passively or reinterpret the negative image as fictional or unrelated. A practice session was included, with 2-minute breaks between blocks. The total task duration was 25 min. CR ability was indexed as the average difference in negative emotion ratings

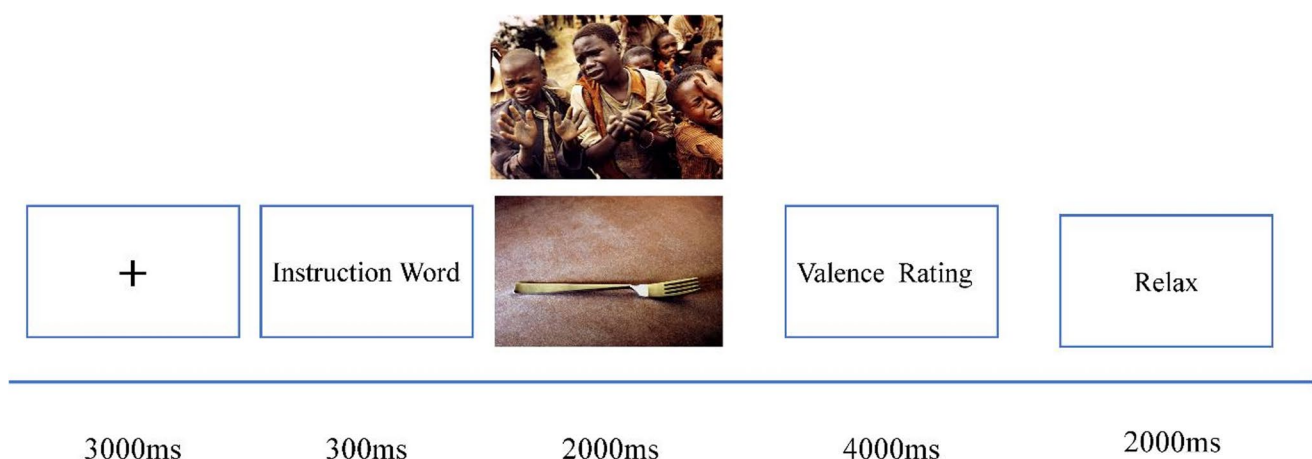


Fig. 1 The flowchart for the emotion regulation task (ERT)

between the watch-negative and reappraise-negative conditions, with higher scores indicating greater reappraisal success. Task flow is illustrated in Fig. 1.

Procedure

All experimental tasks were programmed and administered using E-Prime 2.0 software (Psychology Software Tools). Upon arrival, participants initially completed a set of self-report questionnaires, including the Self-Rating Depression Scale (SDS) and the Beck Anxiety Inventory (BAI). Older adult participants also completed the Mini-Mental State Examination (MMSE; Folstein et al., 1975) as a screening measure of global cognitive status; only those scoring ≥ 25 were eligible to proceed to the experimental phase. Eligible participants subsequently completed three computerized cognitive tasks in a fixed order: the numerical WM task, the CR task, and finally the 1-back WM task. Prior to each task, the experimenter provided standardized verbal instructions and visual examples to ensure adequate task comprehension. Following completion of all procedures, participants were debriefed and compensated financially for their time. The study protocol was approved by the Institutional Review Board of Northwest Normal University, and written informed consent was obtained from all participants before participation.

Analysis Plan

The analytical strategy for Study I was designed to systematically address its primary research questions. First, to examine the effectiveness of the experimental manipulation in the CR task, a one-way repeated-measures analysis of variance (ANOVA) was conducted to compare subjective emotional ratings across the watch-neutral, watch-negative, and reappraise-negative conditions. Second, independent-samples

t-tests were performed to assess age-related differences in performance on the WM updating tasks and the CR task. Third, to investigate the association between WM and CR and the potential moderating role of age, Pearson correlation analyses and hierarchical regression analyses were conducted. Multiple WM tasks were included to provide a comprehensive assessment of the WM updating construct. In addition, the regression analyses examined whether age moderated the relationship between WM updating ability and CR performance.

Results

Manipulation test

A repeated-measures analysis of variance (ANOVA) revealed a significant main effect of the experimental condition ($F(2, 642) = 607.81, p < .001, \eta_p^2 = 0.65$). Negativity scores in the watch-neutral condition ($M = 1.66, 95\% \text{ CI} = (4.79, 5.37)$) were significantly lower than those in both the reappraise-negative condition ($M = 5.08, 95\% \text{ CI} = (1.53, 1.79), p < .001$) and the watch-negative condition ($M = 7.07, 95\% \text{ CI} = (6.86, 7.27), p < .001$). Additionally, scores in the reappraise-negative condition were significantly lower than those in the watch-negative condition ($p < .001$), confirming the effectiveness of the experimental manipulations.

Descriptive Statistics

Table 1 presents the descriptive statistics for participants' accuracy on the WM tasks (numerical WM and 1-back) and the CR task for both the young women and older women. Independent-samples *t* tests revealed significant performance differences between the two groups on the numerical WM tasks (both "easy" and "difficult"), 1-back and the

Table 1 Performance of tasks measured for the young and older women groups in Study I

Task		Young women (<i>n</i> = 101) <i>M</i> ± <i>SD</i>	Older women (<i>n</i> = 114) <i>M</i> ± <i>SD</i>
Numerical WM	“Easy” task accuracy	0.89 ± 0.12	0.60 ± 0.20
	“Difficult” task accuracy	0.86 ± 0.15	0.55 ± 0.21
1-back	Accuracy	0.82 ± 0.16	0.75 ± 0.16
CR task	“Reappraise negative” accuracy – “watch negative” accuracy	2.74 ± 1.93	1.32 ± 1.99

CR task. Specifically, for the numerical WM “easy” task, the mean accuracy of older women was lower than that of young women ($t(213) = 13.29, p < .001$, Cohen’s $d = 1.73$). Similarly, for the numerical WM “difficult” task, older women performed worse, ($t(213) = 12.46, p < .001$, Cohen’s $d = 1.68$). And for the 1-back task, the mean accuracy of older women was lower than that of young women ($t(213) = 2.99, p < .01$, Cohen’s $d = 0.44$). On the CR task, older women scored lower than young women, ($t(213) = 5.32, p < .001$, Cohen’s $d = 0.72$). No significant group differences were found on the 1-back task.

Correlation Analysis

Pearson correlation analysis was conducted to explore the relationship between WM and CR, controlling for years of education, anxiety, and depression. The score on the numerical WM “easy” task was significantly positively correlated with the score on the numerical WM “difficult” task ($r = .72, p < .001$) and with the score on the CR task ($r = .29, p < .001$). The performance on the CR task was significantly positively correlated with the performance on the numerical WM “difficult” task ($r = .25, p < .001$) and with the performance on the 1-back task ($r = .16, p < .05$). The remaining correlation coefficients among the variables (as shown in **Table S1**) were not statistically significant.

Hierarchical Linear Regression Model

To examine the moderating effect of age on the relationship between WM and CR, multiple linear regression analyses were conducted, controlling for years of education, depression, and anxiety. Correlation analyses indicated that performance on the 1-back task was not significantly correlated with CR scores. Therefore, the 1-back task was not included as a primary predictor in the moderation analyses. The numerical WM task, which demonstrated a significant correlation with CR, served as the key predictor to test the WM–CR relationship and its moderation by age.

The scores on the CR task (reappraise-negative minus watch-negative) were used as dependent variables in a moderated multiple regression analysis (using the enter method), with numerical WM “easy” task accuracy, group (dummy variable; 0 = young women, 1 = older women), and the interaction term (numerical WM “easy” task × group) as predictors (see **Table 2** for details). Model 3 showed that the interaction between numerical WM “easy” task accuracy and group significantly predicted CR ($B = -4.467, SE = 1.874, p < .05, 95\% CI (-8.161, -0.772), R^2 = 0.167$), suggesting that age moderates the relationship between the WM “easy” task and CR performance. Separate regression analyses for each group revealed that the numerical WM “easy” task score significantly predicted the CR score for young women (simple slope = 4.21, $t = 4.61, p < .001$), but

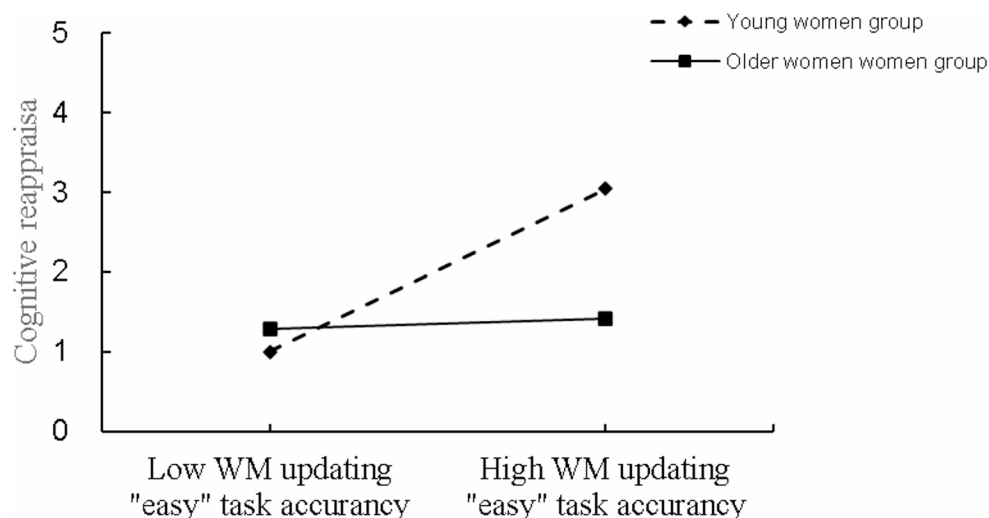
Table 2 Hierarchical regression with the numerical WM “easy” task × group on the CR task as the criterion in Study I

Predictor	B	SE	<i>p</i>	<i>R</i> ²	Δ <i>R</i> ²	F
Model 1			0.224	0.020		1.399
Intercept	−0.307	1.130	0.786			
Years of education	0.080	0.060	0.181			
Depression	0.279	1.952	0.887			
Anxiety	0.051	0.043	0.238			
Model 2^a			0.000	0.145	0.125	7.072
Intercept	2.273	1.158	0.051			
Numerical WM “easy” task	1.345	0.808	0.098			
Group	−1.291	0.386	0.001			6.971
Model 3^a			0.000	0.167	0.023	
Intercept	2.122	1.147	0.066			
Numerical WM “easy” task	1.554	0.804	0.055			
Group	−0.647	0.468	0.168			
Numerical WM “easy” task × group	−4.467	1.874	0.018			

Group (age category) was coded 0 for the young women group and 1 for the older women group

^a Models included years of education, depression, and anxiety. Significant effects are displayed in bold text

Fig. 2 Association between numerical WM “easy” task performance and cognitive reappraisal task scores. All scores are mean centralized



not for older women (simple slope = -0.50 , $t = -0.67$, $p = .50$), as shown in Fig. 2. Similarly, the interaction between numerical WM “difficult” task accuracy and group did not predict CR ($B = -1.805$, $SE = 1.568$, $p = .251$) (see Table S2 for details).

Study II

Building on the cross-sectional findings from Study I, Study II employed a randomized controlled design to directly test the causal efficacy of WMT in enhancing cognitive reappraisal (CR) ability. Specifically, this experiment tested the central hypothesis that targeted enhancement of working memory (WM) updating efficiency—a core cognitive resource for CR—would lead to measurable improvements in CR performance among older women. The sample for Study II consisted exclusively of community-dwelling older women aged 65–85 years. This focused approach was informed by the results of Study I, which demonstrated significant age-related disparities in WM and CR, thereby providing a clear empirical rationale for examining a targeted cognitive intervention within this population.

Method

Participants

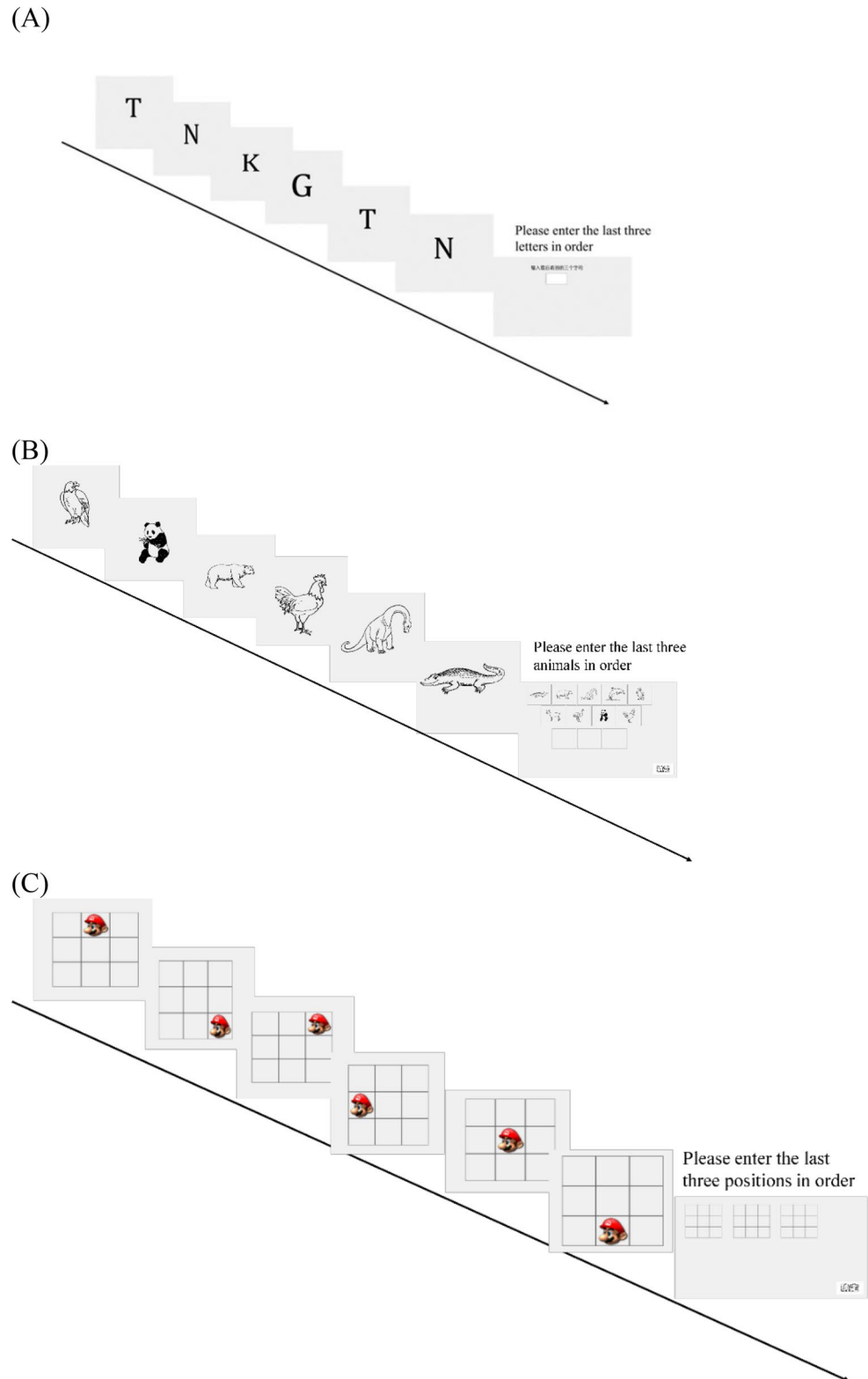
A total of 75 older women were recruited through poster advertisements and screened according to the criteria of Study I. Participants were randomly assigned to one of three groups: Training group, Active control group, and No-contact control group, with 25 participants in each group. During the intervention and assessment phases, participant attrition and exclusion occurred as follows: five participants dropped out for personal reasons (Training group: 2; Active control group: 2; No-contact control group: 1). Additionally, two participants (Training group: 1; Active control group: 1) did

not complete all experimental measures and were therefore excluded from the final analysis. Consequently, the final analyzed sample comprised 68 participants: the Training group ($n = 22$), the Active control group ($n = 22$), and the No-contact control group ($n = 24$). The training group consisted of 22 participants ($M = 70.64$, $SD = 5.47$) with 7–15 years of education ($M = 11.14$, $SD = 2.03$). The Active control group included 22 participants ($M = 70.77$, $SD = 3.93$) with 2–16 years of education ($M = 10.32$, $SD = 2.63$). The No-contact control group consisted of 24 participants ($M = 72.63$, $SD = 5.72$) with 6–16 years of education ($M = 10.33$, $SD = 2.02$). One-way analyses of variance (ANOVAs) confirmed that the three groups did not differ significantly at baseline in terms of age ($F(2, 65) = 1.10$, $p = .305$), years of education ($F(2, 65) = 4.89$, $p = .381$) across the groups. All participants provided written informed consent prior to enrollment and were compensated upon study completion. The research protocol received full approval from the Ethics Committee of Northwest Normal University and was conducted in strict adherence to the approved guidelines and ethical principles.

Interventions

Adaptive Running Memory Task The adaptive training regimen was administered over 20 consecutive days. The core task involved an adaptive running memory paradigm, adapted from previous work (Xiu et al., 2016; Zhao et al., 2023). This task required participants to continuously update working memory contents by recalling the last three items of a sequentially presented, unpredictable-length sequence. To foster the training of a domain-general updating process, the paradigm featured three distinct content variants in each session: letters, animals, and spatial positions within a nine-square grid (see Fig. 3 for a schematic). In the letter variant, for example, a sequence of capital letters (varying in length from 5 to 11 items) was presented centrally, one at a time.

Fig. 3 The schematic diagram of active training (A) Working memory training (WMT)-letters subtasks. (B) Working memory training (WMT)- animals subtasks. (C) Working memory training (WMT)- nine squares' subtasks



Participants were required to verbally recall the last three letters in their order of presentation.

A core feature of the training was its trial-by-trial adaptive algorithm, which dynamically adjusted task difficulty

(operationalized as stimulus presentation duration) to match each participant's performance. The initial presentation duration was set at 1750 ms. Performance was evaluated at the end of each block (comprising five trials). If a participant

correctly recalled the target items on three or more trials within a block, the presentation duration for the subsequent block was shortened by 100 ms, thereby increasing cognitive load. Conversely, if performance fell below this threshold, the duration was lengthened by 100 ms to reduce task difficulty. This adjustment rule was applied independently to each task variant, and the adjusted duration parameters were carried forward to the following day's session, ensuring a continuously personalized and challenging training trajectory. The animal and nine-square variants followed the same procedural and adaptive logic.

Active Control Task Participants assigned to the Active Control group completed a fixed-difficulty, non-adaptive version of the running memory task. This version was structurally identical to the adaptive training task but was specifically designed to eliminate the adaptive component through two critical modifications: (1) all sequences were fixed at a length of five items, and (2) stimulus presentation duration was held constant at 750 ms—a duration empirically associated with high task demand based on baseline assessments. This control condition accounted for potential non-specific effects of repeated task exposure, computer-based interaction, and general task engagement, while isolating the contribution of the performance-contingent, adaptive difficulty progression that defined the active training regimen.

Pre- and Post-Training Tasks The CR task and WM tasks were the same as in Study I.

Procedure

During the initial laboratory session, all participants provided written informed consent and completed a demographic questionnaire, the Self-Rating Depression Scale (SDS), and the Beck Anxiety Inventory (BAI). They then performed the baseline cognitive assessment, which included the numerical working WM task (administered in both easy and difficult conditions), the 1-back task, and the CR task, with a brief recess between assessments.

For participants in the Training group and Active Control group, the intervention phase commenced 1–3 days after the baseline session. These participants completed one training session per day for 20 consecutive days. Each daily session involved all three variants (letters, animals, and nine-square) of their respective running memory task (adaptive for the Training group and fixed-difficulty for the Active Control group). The order of the three variants was counterbalanced across sessions, but participants had the flexibility to complete them in a self-preferred order within a single session. Session duration varied due to the adaptive nature of the Training group, typically ranging between 20 and 30 min.

Participants in the No-Contact Control group received no intervention during this period. The post-training assessment was administered 1–3 days after the final (20th) training session, resulting in a 21–23 day interval between pre- and post-test measurements across all three groups. This follow-up assessment mirrored the baseline assessment and included the numerical WM tasks, the 1-back task, and the CR task.

Analysis Plan

The analyses for Study II were designed to evaluate the efficacy of the WMT intervention. First, manipulation checks for the CR task were conducted at pre- and post-training using repeated-measures ANOVAs to ensure baseline equivalence and task sensitivity. Second, to examine the effects of training on WM updating and CR performance, mixed-design ANOVAs were performed on the numerical WM tasks (easy and difficult conditions), the 1-back task, and the CR task. In these analyses, Groups (Training group, Active Control, No-Contact Control) served as a between-subjects factor, and Time (pre- vs. post-training) served as a within-subjects factor. We hypothesized that participants in the Training group would exhibit greater performance gains compared to both the Active Control and No-Contact Control groups.

Results

Manipulation test

To evaluate the validity of the CR task pre- and post-training, scores were analyzed using repeated-measures ANOVA. Before training, a significant main effect of experimental condition was observed ($F(2, 201) = 330.551, p < .001, \eta_p^2 = 0.77$). Scores in the “watch neutral” condition were significantly lower than those in both the “reappraisal negative” and “watch negative” conditions. Additionally, scores in the “reappraisal negative” condition were significantly lower than those in the “watch negative” condition. Similarly, post-training analyses revealed a significant main effect of condition ($F(2, 201) = 548.882, p < .001, \eta_p^2 = 0.69$), with scores in the “watch neutral” condition significantly lower than those in the “reappraisal negative” and “watch negative” conditions, and scores in the “reappraisal negative” condition lower than those in the “watch negative” condition ($p < .001$). These results indicate the experimental manipulation was effective.

Training Effect

Across the 20-day period, average task difficulty levels for the three WMT subtasks—letters, animals, and

nine-square—were analyzed in both the Training and Active control groups. In the letters task, average difficulty in the Training group improved from 1715.15 ms to 486.36 ms, while the Active Control group improved from 1681.80 ms to 756.82 ms. In the animals task, the Training group improved from 1913.71 ms to 518.18 ms, and the Active Control group from 1783.99 ms to 753.03 ms. In the nine-square task, difficulty in the Training group increased from 1819.70 ms to 500.00 ms, and in the Active Control group from 1681.03 ms to 750.00 ms. Notably, the average difficulty of the final training session was significantly higher than that of the first session for both groups across all tasks (see Figure S1 for details).

Transfer Effect

Table 3 presents the pre- and post-training scores on the 1-back accuracy, numerical WM task accuracy, and CR task, along with change scores. As shown in Table 4, no significant interaction effect was found for the numerical WM “easy” task or 1-back performance. However, a significant interaction between time and group emerged for the numerical WM “difficult” task accuracy ($F(2, 65) = 7.17, p < .01, \eta_p^2 = 0.18$). Simple effect analyses indicated that, in the training group, post-training accuracy ($M = 0.84, 95\% \text{ CI} = (0.76, 0.92)$) was significantly higher than pre-training accuracy ($M = 0.55, 95\% \text{ CI} = (0.46, 0.65), p < .001$). In the Active Control group, post-training accuracy ($M = 0.67, 95\% \text{ CI} = (0.59, 0.75)$) was also higher than pre-training ($M = 0.57, 95\% \text{ CI} (0.47, 0.66), p < .05$), while no significant difference was found in the No-contact control group ($p = .42$). Post-training accuracy in the Training group was significantly greater than in both the sham ($p < .05$) and No-contact control groups ($p < .001$), whereas no significant difference was found between the sham and No-contact control groups ($p = .03$; see Fig. 4 and Appendix Part 6 for details).

For CR task scores, a significant time $\times$ group interaction was also observed ($F(2, 65) = 6.05, p < .05, \eta_p^2 = 0.16$). In the Training group, post-training scores ($M = 4.53, 95\% \text{ CI} (3.69, 5.38)$) were significantly higher than pre-training scores ($M = 1.65, 95\% \text{ CI} (0.78, 0.51)$). No significant pre–post changes were found in the sham ($p = .33$) or control ($p = .38$) groups. Post-training CR scores in the Training group were significantly greater than those in the sham ($M = 1.68, 95\% \text{ CI} (0.83, 2.52), p < .001$) and No-contact control groups ($M = 1.20, 95\% \text{ CI} (0.39, 2.01), p < .001$) (see Fig. 5).

Table 3 Tasks measured at each time point for the three groups in Study II

Numerical WM “easy” task accuracy (%)	Training group ($n = 22$) $M \pm SD$	Active control group ($n = 22$) $M \pm SD$	No-contact control group ($n = 24$) $M \pm SD$
Pre-training	0.62 $\pm$ 0.18	0.56 $\pm$ 0.23	0.52 $\pm$ 0.22
Post-training	0.85 $\pm$ 0.18	0.71 $\pm$ 0.14	0.61 $\pm$ 0.23
Improvement	0.23	0.15	0.09
Numerical WM “difficult” task accuracy (%)			
Pre-training	0.55 $\pm$ 0.21	0.57 $\pm$ 0.22	0.49 $\pm$ 0.23
Post-training	0.84 $\pm$ 0.19	0.67 $\pm$ 0.14	0.52 $\pm$ 0.22
Improvement	0.29	0.10	0.03
1-back accuracy (%)			
Pre-training	0.68 $\pm$ 0.13	0.77 $\pm$ 0.20	0.73 $\pm$ 0.18
Post-training	0.72 $\pm$ 0.20	0.83 $\pm$ 0.11	0.76 $\pm$ 0.20
Improvement	0.04	0.06	0.03
CR task scores			
Pre-training	1.65 $\pm$ 2.23	1.13 $\pm$ 2.16	0.73 $\pm$ 1.59
Post-training	4.53 $\pm$ 2.27	1.68 $\pm$ 1.77	1.20 $\pm$ 1.89
Improvement	2.88	0.55	0.47

M = mean, SD = standard deviation

Table 4 Groups and time repeated-measures ANOVA ($n = 68$) in Study II

Tasks	Variables	F	p	η_p^2
Numerical WM “easy” task accuracy (%)	Time	34.46	<0.001	0.35
	Group	5.76	<0.01	0.15
	Time $\times$ group	2.50	0.090	0.07
Numerical WM “difficult” task accuracy (%)	Time	26.01	<0.001	0.29
	Group	7.72	<0.01	0.19
	Time $\times$ group	7.17	<0.01	0.18
1-back accuracy (%)	Time	3.29	0.07	0.05
	Group	2.66	0.07	0.08
	Time $\times$ group	0.02	0.98	0.00
CR task scores	Time	16.60	<0.001	0.20
	Group	12.54	<0.001	0.28
	Time $\times$ group	6.05	<0.01	0.16

Integrating Evidence Across Studies

The contrasting findings across studies clarify the nature of the absent baseline WM–CR correlation in older women. While no association was observed at baseline (Study I), WMT in Study II not only improved both WM updating and CR performance but also elicited strong post-training correlations between gains in these domains (e.g., for the easy WM task: $r = .88, p < .01$; see Table S3)). Furthermore, post-training WM performance significantly predicted CR scores in regression models (all $p < .001$; see Table S4).

Fig. 4 Numerical WM “difficult” task accuracy in pre-training and post-training for three groups. Error bars represent standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

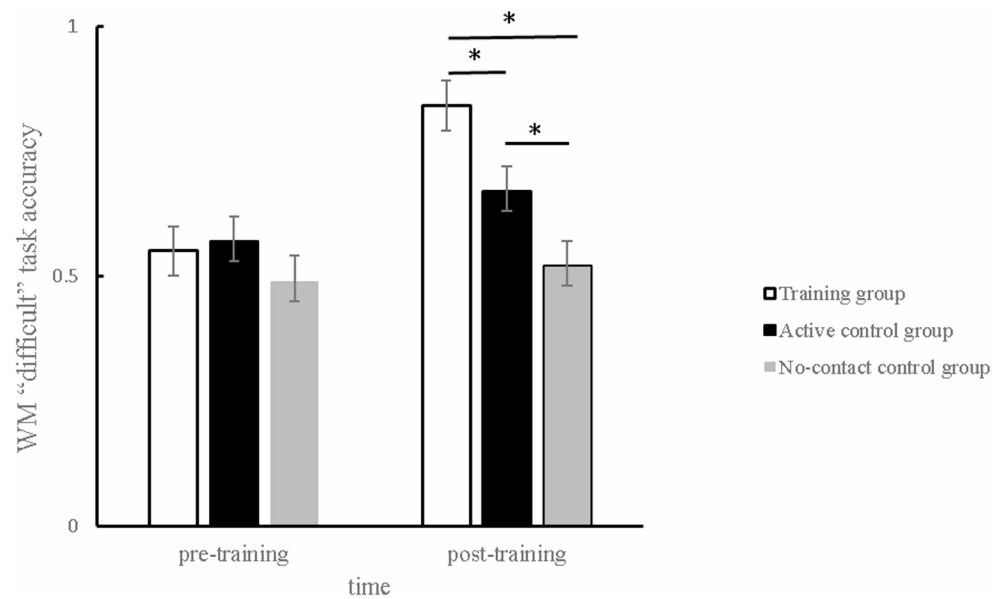
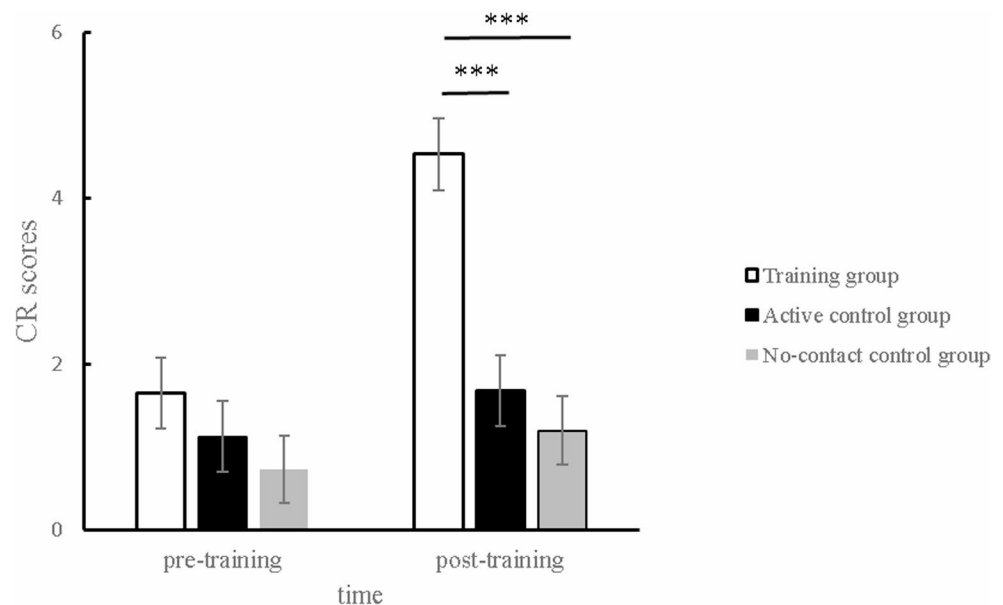


Fig. 5 CR task scores in pre-training and post-training for three groups. Error bars represent standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$



This pattern indicates that the initial null correlation likely reflected a modifiable state of cognitive resource limitation.

General Discussion

The present research yielded two principal findings. First, at baseline, older women exhibited significantly lower WM performance than young women, as reflected in reduced accuracy on both simple and difficult numerical WM tasks, a pattern consistent with prior evidence of age-related cognitive decline (Nissim et al., 2016). Notably, although WM updating was significantly associated with CR ability in young women, this relationship was absent in older women. Second, participation in an WMT protocol led to

significant improvements in both WM and CR performance among older women, accompanied by the emergence of a strong post-training correlation between gains in these two domains.

The cross-sectional finding of a null association between WM and CR in older women at baseline warrants careful interpretation. While a statistical floor effect represents one plausible explanation, we propose a more specific cognitive account centered on an efficiency bottleneck, derived from the “necessary-but-not-sufficient” role of WM updating in CR (Growney & English, 2023). According to this framework, age-related reductions in WM efficiency may fall below a functional threshold required to provide observable behavioral support in a standard CR task, thereby masking

their relationship at baseline (Martins et al. 2018a; 2018b). Importantly, the experimental findings from Study II substantiate this interpretation. The observation that WMT not only enhanced WM and CR performance but also produced robust post-training correlations between WM gains and CR improvement indicates that elevating WM efficiency beyond this threshold effectively frees up its functional contribution to CR (Opitz et al., 2012; Growney & English, 2023). Consequently, the initial dissociation between WM and CR in older women likely reflects a modifiable state of resource insufficiency rather than a fundamental decoupling of these cognitive processes in later adulthood.

This pattern aligns with theoretical models of aging, particularly the Selective Optimization with Compensation (SOC) model as applied to emotion regulation (Urry & Gross, 2010). The observed decline in WM updating and its correlation with CR in young women, but not older women at baseline, resonates with the SOC model, which suggests that older adults may adapt to cognitive constraints by optimizing available resources or relying on alternative strategies. The initial absence of a WM-CR link in older women may reflect an adaptation phase, where WM resources have declined below a critical threshold required for efficient CR, triggering a shift in strategy use. Our training intervention, by enhancing WM efficiency, effectively provided the requisite “cognitive resource” for optimization. Furthermore, the older group demonstrated lower CR ability than young adults, consistent with previous findings that older adults can be less effective at using reappraisal to regulate negative emotions (Opitz et al., 2012). However, this finding contrasts with studies reporting no age-related differences in CR when using moderately negative stimuli (Opitz et al., 2014), a discrepancy that may be due to methodological differences between studies.

The integration of findings from both studies supports the conclusion that WM updating ability is associated with CR ability. Although Study I showed that only young women’s WM performance significantly predicted their CR scores, Study II revealed a significant correlation following training. This convergence is consistent with prior evidence linking WM to emotion regulation (Hendricks & Buchanan, 2016; Sperduti et al., 2017; Toh & Yang, 2022). Specifically, WM has been linked to the regulation of negative emotions through reappraisal (Hendricks & Buchanan, 2016), with associations particularly evident for high-intensity stimuli (Sperduti et al., 2017), and WM can also influence the impact of CR on well-being (Toh & Yang, 2022). This provides evidence that WM serves as a key cognitive resource for successful emotion regulation in late adulthood. Prior research indicates that both WM ability (Nickerson et al., 2017) and emotion regulation (Eldesouky & English, 2018) decline with age, with neuroimaging studies showing that WM impairments involve the DLPFC (Nickerson et al., 2017), whereas CR

deficits involve prefrontal regions, the cingulate gyrus, and the amygdala (Ochsner et al., 2012).

The successful transfer of WMT to CR in our study contrasts with some prior null findings (e.g., Vanderhasselt et al., 2021; Freidle et al., 2020). This discrepancy may be explained by key methodological differences. First, our training was highly adaptive, intensive, and targeted WM updating through multiple variants. Second, our focus on older women, a group with particular vulnerability and sex-specific neural correlates of CR (McRae et al., 2008), may have identified a population with both need and latent capacity. Third, our process-based CR task may be more sensitive to training effects than self-report measures. Therefore, WMT efficacy for emotion regulation may be moderated by training intensity, adaptivity, population characteristics, and measurement specificity.

An intriguing finding is the differential sensitivity exhibited by the two WM updating tasks. In Study I, only the numerical WM task demonstrated a significant age-moderating effect on the WM-CR relationship. In Study II, significant training effects were observed for the challenging numerical WM task and CR, but not for the 1-back task. This dissociation suggests that the tasks engage distinct cognitive demands that are differentially related to CR. The numerical WM task necessitates active recall, maintenance, and sequential updating of items, heavily relying on the monitoring and revision of WM representations—an essential updating function (Zhu & Zhao, 2023) analogous to the operations required for generating and maintaining alternative appraisals in CR (Ochsner et al., 2012). The high process similarity between this task and our adaptive training explains the observed near-transfer (Zhao et al., 2023). In contrast, the 1-back task, while involving WM, is relatively simple and emphasizes recognition (Du et al., 2018; Oberauer, 2005); furthermore, a more challenging n-back task (e.g., 2-back) may better distinguish individual WM updating (Blacker et al., 2017; Oberauer, 2005; Zhao et al., 2023). Thus, the numerical WM task serves as a more sensitive and theoretically relevant measure of cognitive control processes that are essential for effective reappraisal.

What accounts for the lack of improvement in CR among the Active Control Group? The absence of improvement is consistent with our theoretical model. This group engaged in a fixed-difficulty task that controlled for non-specific factors but lacked the adaptive, performance-contingent escalation of difficulty. We propose that the active ingredient is the systematic training of WM updating efficiency. The adaptive protocol was specifically designed to enhance this efficiency beyond the functional threshold required to support reappraisal. The fixed-difficulty task did not offer this progressive challenge and, consequently, did not sufficiently enhance the specific cognitive resource (WM updating efficiency) necessary for achieving CR gains (Long et al., 2024).

It is essential to contextualize our findings within the broader discourse on cognitive training efficacy. A substantial body of literature raises valid concerns regarding limited transfer effects, arguing that improvements are often task-specific (e.g., Himi et al., 2023; Ripp et al., 2022). However, our study demonstrated a transfer effect to emotion regulation. Several factors may account for this observation: our highly adaptive, multi-domain protocol aimed at strengthening domain-general WM; our sample of older women, who exhibited pronounced declines, had substantial “room for improvement”; and the transfer to CR is theoretically plausible given the overlapping neural substrates (e.g., DLPFC) and cognitive resources (e.g., executive control) shared by WM and CR tasks (Ochsner et al., 2012). Therefore, while acknowledging general skepticism, our results suggest that meaningful transfer can be achieved under specific conditions—namely, adaptive, multi-domain training targeted at theoretically linked abilities within a vulnerable population.

Several limitations of this study should be acknowledged. First, the long-term effects of WMT remain unclear due to the absence of longitudinal follow-up, although one study has reported effects persisting for up to three months (Long et al., 2024). Second, emotion regulation was assessed exclusively via self-report measures, which are susceptible to reporting biases (Li et al., 2019); incorporating physiological indices such as heart rate variability may improve measurement precision (Koenig et al., 2016; Xiu et al., 2016). Third, WM gains were assessed using tasks that were structurally similar to the training paradigm, indicating robust near-transfer; future research should incorporate more structurally distinct WM measures to better isolate improvements in the underlying cognitive construct. Fourth, the adaptive protocol primarily adjusted task difficulty along the dimension of processing speed; future studies could explore adaptive algorithms targeting additional parameters to further enhance training efficacy. Fifth, the present study focused on downregulating negative emotions via dissociative reappraisal; future work should examine a broader range of CR strategies and regulatory goals (McRae et al., 2012; Vlasenko et al., 2024).

Several limitations of this study should be acknowledged. First, the long-term effects of WMT remain unclear due to the absence of longitudinal follow-up, although one study has reported effects persisting for up to three months (Long et al., 2024). Second, emotion regulation was assessed exclusively via self-report measures, which are susceptible to reporting biases (Li et al., 2019); incorporating physiological indices such as heart rate variability may improve measurement precision (Koenig et al., 2016; Xiu et al., 2016). Third, WM gains were assessed using tasks that were structurally similar to the training paradigm, indicating robust near-transfer; future research should incorporate more structurally distinct

WM measures to better isolate improvements in the underlying cognitive construct. Fourth, the adaptive protocol primarily adjusted task difficulty along the dimension of processing speed; future studies could explore adaptive algorithms targeting additional parameters to further enhance training efficacy. Fifth, the present study focused on downregulating negative emotions via dissociative reappraisal; future work should examine a broader range of CR strategies and regulatory goals (McRae et al., 2012; Vlasenko et al., 2024).

Furthermore, emotion regulation ability and regulation frequency represent distinct constructs (Cohen & Mor, 2018); although older adults may engage in regulation more frequently, their regulatory effectiveness may be reduced (Urry & Gross, 2010). Investigating the role of WM in both dimensions could inform more comprehensive models of emotion regulation. Finally, the sample consisted exclusively of older women; the absence of a WM–CR association in older men (Zhao et al., 2023) remains unexplained, warranting further investigation into gender differences and the development of interventions for more diverse populations. Future research should examine training durability through long-term follow-ups, investigate underlying neural mechanisms, assess benefits in older men and clinical populations (e.g., mild cognitive impairment), and develop “emotional WMT” paradigms that integrate affective stimuli to enhance real-world transfer (Pan et al., 2024).

In conclusion, by integrating cross-sectional and experimental approaches, the present research demonstrates that the absence of a WM–CR correlation in older women reflects a reversible efficiency bottleneck rather than a permanent decoupling. Adaptive training targeting WM updating can remediate this resource insufficiency, thereby restoring its functional linkage to emotion regulation. Collectively, these findings underscore the latent plasticity of cognitive–emotional interactions in aging and highlight the potential of targeted cognitive training to support emotional health in late adulthood.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s41465-026-00354-8>.

Acknowledgements We are grateful to Dr. Cohen from the University of Haifa for her insightful comments on the manuscript and the data analysis.

Authors' contributions Q.C.(Qing Chai) conceived the study, performed experiments and collected data. Y. G (Yue Gao) analyzed data, and wrote the original manuscript. Y.G.(Yajie Gong)suggested revisions to the preliminary manuscript.X.Z.(Xin Zhao) acquired funding and critically revised the manuscript. All authors approved the final version.

Funding This work was supported by the National Natural Science Foundation of China (32260207) and the Humanities and Social Sciences Research Project of the Ministry of Education of China (21XJA190005).

Data Availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate This human study was reviewed and approved by the Ethics Committee of Northwest Normal University and was performed in accordance with ethical standards of the American Psychological Association. Informed consent was obtained from all individual participants included in the study.

Competing interests The authors declare no competing interests.

References

- Blacker, K. J., Negoita, S., Ewen, J. B., & Courtney, S. M. (2017). N-back versus complex span working memory training. *Journal of Cognitive Enhancement*, 1(4), 434–454. <https://doi.org/10.1007/s41465-017-0044-1>
- Broadway, J. M., & Engle, R. W. (2010). Validating running memory span: Measurement of working memory capacity and links with fluid intelligence. *Behavior Research Methods*, 42(2), 563–570. <https://doi.org/10.3758/BRM.42.2.563>
- Charles, S. T., & Carstensen, L. L. (2008). Unpleasant situations elicit different emotional responses in younger and older adults. *Psychology and Aging*, 23(3), 495–504. <https://doi.org/10.1037/a0013284>
- Cohen, N., & Mor, N. (2018). Enhancing reappraisal by linking cognitive control and emotion. *Clinical Psychological Science*, 6(1), 155–163. <https://doi.org/10.1177/2167702617731379>
- Cohen, N., & Ochsner, K. N. (2018). From surviving to thriving in the face of threats: The emerging science of emotion regulation training. *Current Opinion in Behavioral Sciences*, 24, 143–155. <https://doi.org/10.1016/j.cobeha.2018.08.007>
- Diestel, S., & Schmidt, K. H. (2011). The moderating role of cognitive control deficits in the link from emotional dissonance to burnout symptoms and absenteeism. *Journal of occupational health psychology*, 16(3), 313–330. <https://doi.org/10.1037/a0022934>
- Du, X., Ji, Y., Chen, T., Tang, Y., & Han, B. (2018). Can working memory capacity be expanded by boosting working memory updating efficiency in older adults? *Psychology and Aging*, 33(8), 1134–1151. <https://doi.org/10.1037/pag0000311>
- Eldesouky, L., & English, T. (2018). Another year older, another year wiser? Emotion regulation strategy selection and flexibility across adulthood. *Psychology and Aging*, 33(4), 572–585. <https://doi.org/10.1037/pag0000251>
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2), 175–191. <https://doi.org/10.3758/brm.39.2.175>
- Folstein, M. F., Folstein, S. E., & McHugh, P. R. (1975). 'Mini-mental state': A practical method for grading the cognitive state of patients for the clinician. *Journal of Psychiatric Research*, 12(3), 189–198. [https://doi.org/10.1016/0022-3956\(75\)90026-6](https://doi.org/10.1016/0022-3956(75)90026-6)
- Freidle, M., Nilsson, J., Lebedev, A. V., & Lövdén, M. (2020). No evidence for any effect of multiple sessions of frontal transcranial direct stimulation on mood in healthy older adults. *Neuropsychologia*, 137, 107325. <https://doi.org/10.1016/j.neuropsychologia.2019.107325>
- Gross, J. J. (1998). The emerging field of emotion regulation: An integrative review. *Review of General Psychology*, 2(3), 271–299. <https://doi.org/10.1037/1089-2680.2.3.271>
- Growney, C. M., & English, T. (2023). Fluid and crystallized cognitive resources differentially linked to emotion regulation success in adulthood. *Emotion (Washington D C)*, 23(2), 589–594. <https://doi.org/10.1037/emo0001087>
- Hendricks, M. A., & Buchanan, T. W. (2016). Individual differences in cognitive control processes and their relationship to emotion regulation. *Cognition & emotion*, 30(5), 912–924. <https://doi.org/10.1080/02699931.2015.1032893>
- Himi, S. A., Stadler, M., von Bastian, C. C., Bühner, M., & Hilbert, S. (2023). Limits of near transfer: Content- and operation-specific effects of working memory training. *Journal of experimental psychology General*, 152(5), 1305–1333. <https://doi.org/10.1037/xge0001328>
- Holden, C. (2005). Sex and the suffering brain. *Science*, 308(5728), 1574. <https://doi.org/10.1126/science.308.5728.1574>
- Koenig, J., Kemp, A. H., Beauchaine, T. P., Thayer, J. F., & Kaess, M. (2016). Depression and resting state heart rate variability in children and adolescents — A systematic review and meta-analysis. *Clinical Psychology Review*, 46, 136–150. <https://doi.org/10.1016/j.cpr.2016.04.013>
- Lang, P. J., Bradley, M. M., & Cuthbert, B. N. (1999). *International affective picture system (IAPS): Technical manual and affective ratings*. The Center for Research in Psychophysiology, University of Florida.
- Levens, S. M., & Gotlib, I. H. (2010). Updating positive and negative stimuli in working memory in depression. *Journal of Experimental Psychology: General*, 139(4), 654.
- Li, H. X., Long, Q. S., Chen, A. T., & Li, Q. (2019). The influence of reward motivation on emotion regulation. *Sheng Li Xue Bao: (Acta Physiologica Sinica)*, 71(4), 562–574. <https://doi.org/10.13294/j.aps.2019.0044>
- Long, F., Zhou, W., Zhang, T., Zhang, Z., & Zhou, R. (2024). Effects of working memory training on different goals of cognitive reappraisal. *Emotion*, 24(1), 150–163. <https://doi.org/10.1037/emo0001253>
- Lu, B., Hui, M., & Yu-Xia, H. (2005). The development of native Chinese Affective Picture System—A pretest in 46 college students. *Chinese Mental Health Journal*, 19(11), 719–722. <https://doi.org/10.3321/J.ISSN:1000-6729.2005.11.001>
- Martins, B., Florjanczyk, J., Jackson, N. J., Gatz, M., & Mather, M. (2018a). Age differences in emotion regulation effort: Pupil response distinguishes reappraisal and distraction for older but not younger adults. *Psychology and aging*, 33(2), 338–349.
- Martins, B., Sheppes, G., Gross, J. J., & Mather, M. (2018b). Age differences in emotion regulation choice: Older adults use distraction less than younger adults in high-intensity positive contexts. *The Journals of Gerontology: Series B: Psychological Sciences and Social Sciences*, 73(4), 603–611.
- McNab, F., Varrone, A., Farde, L., Jucaite, A., Bystritsky, P., Forssberg, H., & Klingberg, T. (2009). Changes in cortical dopamine D1 receptor binding associated with cognitive training. *Science*, 323(5915), 800–802. <https://doi.org/10.1126/science.1166102>
- McRae, K., Jacobs, S. E., Ray, R. D., John, O. P., & Gross, J. J. (2012). Individual differences in reappraisal ability: Links to reappraisal frequency, well-being, and cognitive control. *Journal of Research in Personality*, 46(1), 2–7. <https://doi.org/10.1016/j.jrp.2011.10.003>
- McRae, K., Ochsner, K. N., Mauss, I. B., Gabrieli, J. J. D., & Gross, J. J. (2008). Gender differences in emotion regulation: An fMRI study of cognitive reappraisal. *Group Processes & Intergroup Relations*, 11(2), 143–162. <https://doi.org/10.1177/1368430207088035>
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex Frontal Lobe tasks: a latent variable analysis. *Cognitive Psychology*, 41(1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>
- Nickerson, A., Garber, B. D., Liddell, B. J., Litz, B. T., Hofmann, S. G., Asnaani, A., Ahmed, O. M., Cheung, J., Huynh, L., Pajak,

- R., & Bryant, R. A. (2017). Impact of cognitive reappraisal on negative affect, heart rate, and intrusive memories in traumatized refugees. *Clinical Psychological Science*, 5, 497–512. <https://doi.org/10.1177/2167702617690857>
- Nissim, N. R., O'Shea, A. M., Bryant, V., Porges, E. C., Cohen, R., & Woods, A. J. (2016). Frontal structural neural correlates of working memory performance in older adults. *Frontiers in Aging Neuroscience*, 8, 328. <https://doi.org/10.3389/fnagi.2016.00328>
- Oberauer, K. (2005). Binding and inhibition in working memory: Individual and age differences in short-term recognition. *Journal of Experimental Psychology General*, 134(3), 368–387. <https://doi.org/10.1037/0096-3445.134.3.368>
- Ochsner, K. N., Silvers, J. A., & Buhle, J. T. (2012). Functional imaging studies of emotion regulation: A synthetic review and evolving model of the cognitive control of emotion. *Annals of the New York Academy of Sciences*, 1251(1), E1–E24. <https://doi.org/10.1111/j.1749-6632.2012.06751.x>
- Opitz, P. C., Lee, I. A., Gross, J. J., & Urry, H. L. (2014). Fluid cognitive ability is a resource for successful emotion regulation in older and younger adults. *Frontiers in Psychology*, 5, 609. <https://doi.org/10.3389/fpsyg.2014.00609>
- Opitz, P. C., Rauch, L. C., Terry, D. P., & Urry, H. L. (2012). Prefrontal mediation of age differences in cognitive reappraisal. *Neurobiology of Aging*, 33(4), 645–655. <https://doi.org/10.1016/j.neurobiolaging.2010.06.004>
- Pan, M., Liu, X., Gao, X., Fu, Z., Liu, L., Li, H., Wang, Y., & Qian, Q. (2024). Feasibility, acceptability, and preliminary efficacy of the unified protocol for transdiagnostic treatment of emotional disorders in adolescents (UP-A) in China: A pilot study. *Behavior Therapy*, 56(1), 145–161. <https://doi.org/10.1016/j.beth.2024.05.003>
- Peng, W., Luo, W., & Zhou, R. (2019). HRV evidence for the improvement of emotion regulation in university students with depression tendency by working memory training. *Acta Psychologica Sinica*, 51(6), 648–661. <https://doi.org/10.3724/SP.J.1041.2019.00648>
- Ripp, I., Emch, M., Wu, Q., Lizarraga, A., Udale, R., von Bastian, C. C., Koch, K., & Yakushev, I. (2022). Adaptive working memory training does not produce transfer effects in cognition and neuroimaging. *Translational psychiatry*, 12(1), 512. <https://doi.org/10.1038/s41398-022-02272-7>
- Schmeichel, B. J., & Tang, D. (2015). Individual differences in executive functioning and their relationship to emotional processes and responses. *Current Directions in Psychological Science*, 24(2), 93–98. <https://doi.org/10.1177/0963721414555178>
- Sperduti, M., Makowski, D., Arcangeli, M., Wantzen, P., Zalla, T., Lemaire, S., & Piolino, P. (2017). The distinctive role of executive functions in implicit emotion regulation. *Acta Psychologica*, 180, 52–61. <https://doi.org/10.1016/j.actpsy.2017.08.007>
- Szymkowicz, S. M., Taylor, W. D., & Woods, A. J. (2022). Augmenting cognitive training with bifrontal tDCS decreases subclinical depressive symptoms in older adults: Preliminary findings. *Brain stimulation*, 15(5), 1037–1039. <https://doi.org/10.1016/j.brs.2022.07.055>
- Toh, W. X., & Yang, H. (2022). Executive function moderates the effect of reappraisal on life satisfaction: A latent variable analysis. *Emotion*, 22(3), 554–571. <https://doi.org/10.1037/emo0000907>
- Urry, H. L., & Gross, J. J. (2010). Emotion regulation in older age. *Current Directions in Psychological Science*, 19(6), 352–357. <https://doi.org/10.1177/0963721410388395>
- Vanderhasselt, M. A., Demeyer, I., Van Imschoot, L., Hoorelbeke, K., & De Raedt, R. (2021). Cognitive control training in healthy older adults: A proof of concept study on the effects on cognitive functioning, emotion regulation and affect. *Cognitive Therapy and Research*, 45(5), 959–968. <https://doi.org/10.1007/s10608-020-10154-9>
- Vlasenko, V. V., Hayutin, I., Pan, C., Vardakis, J. M., Waugh, C. E., Admon, R., & McRae, K. (2024). How do people use reappraisal? An investigation of selection frequency and affective outcomes of reappraisal tactics. *Emotion*, 24(3), 676–686. <https://doi.org/10.1037/emo0001259>
- Xiu, L., Zhou, R., & Jiang, Y. (2016). Working memory training improves emotion regulation ability: Evidence from HRV. *Physiology and Behavior*, 155, 25–29. <https://doi.org/10.1016/j.physbeh.2015.12.004>
- Yang, J., Li, Y., Si, J., & Zhang, M. (2011). Central executive component affects adults' computational estimation performance. *Psychological Exploration*, 31(4), 321–325. <https://doi.org/10.3969/j.issn.1003-5184.2011.04.007>
- Zhao, X., Cai, Y., & Maes, J. H. R. (2023). Association between working memory updating ability and cognitive reappraisal: An investigation into a modulatory role of the individual's sex. *PsyCh Journal*, 12(2), 185–194. <https://doi.org/10.1002/pchj.628>
- Zhao, X., Fu, J., & Maes, J. H. R. (2019). Training and transfer effects of prospective memory training in 13- to 15-year-old children. *Journal of Experimental Child Psychology*, 179, 227–247. <https://doi.org/10.1016/j.jecp.2018.11.014>
- Zhao, X., Xu, Y., Fu, J., & Maes, J. H. R. (2018). Are training and transfer effects of working memory updating training modulated by achievement motivation? *Memory and Cognition*, 46(3), 398–409. <https://doi.org/10.3758/s13421-017-0773-5>
- Zhu, X., & Zhao, X. (2023). Role of executive function in mathematical ability of children in different grades. *Acta Psychologica Sinica*, 55(5), 696–710. <https://doi.org/10.3724/SP.J.1041.2023.00696>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.